

LEINEFELDE, Germany

WMO: 104490

Lat: 51.3933N

Lon: 10.3122E

Elev: 1159

StdP: 14.09

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.7	12.7	2.8	6.6	9.7	7.5	8.4	14.2	27.8	40.6	25.0	38.0	5.7	90	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	81.1	64.9	77.8	63.5	74.8	62.1	66.7	77.2	64.9	74.7	63.4	72.3	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	89.6	70.7	61.2	84.4	68.7	59.7	79.8	67.0	31.9	77.4	30.5	74.7	29.3	72.6	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	19.8	17.0	DB	6.6	88.1	9.4	2.8	-0.2	90.1	-5.7	91.7	-11.0	93.3	-17.8	95.3	(4)
(5)			WB	6.0	70.6	9.3	2.3	-0.7	72.2	-6.1	73.6	-11.3	74.9	-18.1	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	31.5	30.7	38.2	44.0	52.6	57.5	61.9	62.1	55.7	48.1	38.9	34.4
	DBStd	13.16	9.45	9.38	7.43	6.82	6.58	6.29	5.98	5.85	5.61	6.57	6.94	7.06
	HDD50	2686	572	542	369	203	50	7	0	0	10	113	336	484
	HDD65	6888	1037	961	830	630	385	238	135	130	283	526	784	949
	CDD50	1371	0	1	4	23	130	232	370	374	182	53	2	0
	CDD65	99	0	0	0	0	1	13	40	39	5	1	0	0
	CDH74	826	0	0	0	0	15	117	333	327	33	1	0	0
	CDH80	138	0	0	0	0	0	16	50	72	0	0	0	0
Wind	WSAvg	7.1	9.2	8.3	8.0	7.1	6.4	5.9	5.6	5.4	6.3	7.1	7.7	7.9
Precipitation	PrecAvg	27.2	2.2	1.9	2.0	1.7	2.5	2.5	2.9	2.5	2.3	2.0	2.4	2.4
	PrecMax	37.0	4.5	3.9	5.2	3.0	5.7	4.1	4.7	5.2	5.1	6.3	4.4	5.2
	PrecMin	20.5	0.0	0.6	0.4	0.2	0.6	1.2	0.7	0.9	0.3	0.4	0.0	0.9
	PrecStd	4.3	1.1	1.0	1.2	0.6	1.3	0.7	1.1	1.1	1.2	1.3	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.6	55.0	62.4	68.2	75.8	82.6	84.6	87.2	77.8	69.9	57.4	52.7
		MCWB	46.5	49.2	50.3	53.7	60.3	65.4	66.7	66.2	62.1	59.1	51.1	48.2
	2%	DB	46.9	50.5	55.7	63.8	72.5	77.3	80.9	81.3	73.7	64.1	52.8	49.6
		MCWB	44.3	46.3	48.3	52.3	58.6	63.9	65.4	64.1	61.2	56.7	49.1	46.7
	5%	DB	44.9	45.7	51.1	59.9	68.8	73.5	77.8	77.4	69.2	60.4	50.3	46.8
		MCWB	42.6	42.5	45.8	50.1	56.9	61.5	64.0	63.1	59.8	54.6	47.9	44.7
	10%	DB	42.4	41.8	48.0	55.7	65.4	69.5	74.3	73.7	65.3	57.3	48.0	44.3
		MCWB	40.1	39.0	43.8	47.9	55.5	59.9	62.5	61.7	57.9	52.8	45.7	42.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	50.0	51.2	55.9	63.1	68.6	69.7	69.4	64.6	60.9	52.2	49.0
		MCDB	48.9	54.7	59.3	64.9	72.0	78.7	81.4	80.6	74.0	67.1	55.1	51.1
	2%	WB	44.8	46.4	48.6	53.1	60.2	65.1	67.1	66.1	62.3	57.9	49.8	47.0
		MCDB	46.9	49.6	54.6	61.8	69.9	74.5	77.8	77.0	70.3	62.7	52.1	49.3
	5%	WB	42.7	42.4	46.5	50.8	58.0	62.9	65.0	64.4	60.4	55.4	48.0	44.9
		MCDB	44.7	45.4	50.6	58.2	67.1	71.4	74.8	74.6	67.1	59.0	50.5	46.9
	10%	WB	40.3	39.5	44.2	48.5	56.0	60.9	63.3	62.8	58.7	53.3	45.9	42.4
		MCDB	42.4	41.7	47.5	54.8	64.1	68.1	72.1	71.4	65.0	56.7	48.0	44.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.5	11.4	14.4	16.4	15.5	16.6	16.9	14.0	11.5	8.6	6.9
		MCDBR	8.5	13.4	17.1	20.5	22.1	22.0	22.5	22.7	20.5	16.3	12.3	9.0
		MCWBR	7.7	10.2	11.5	11.7	11.8	11.1	10.8	10.0	10.4	9.9	9.6	8.1
	5% WB	MCDBR	8.6	12.8	15.6	18.6	20.4	19.9	20.8	20.1	17.6	14.4	11.5	8.4
		MCWBR	8.0	10.2	10.9	11.5	11.5	11.0	11.0	9.7	9.7	9.5	9.4	7.8
Clear-Sky Solar Irradiance	taub	0.302	0.325	0.361	0.399	0.402	0.400	0.415	0.408	0.373	0.346	0.325	0.298	
	taud	2.403	2.369	2.299	2.241	2.261	2.287	2.249	2.290	2.373	2.440	2.428	2.397	
	Ebn at Noon	228	253	263	266	270	272	265	260	257	242	216	209	
	Edh at Noon	19	26	33	39	41	40	41	37	31	24	19	16	
All-Sky Solar Radiation	RadAvg	230	449	794	1270	1545	1673	1575	1349	974	566	261	174	
	RadStd	30	63	100	156	177	142	208	130	128	63	35	21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page